Performance Testing

1. Introduction

Performance testing is an essential phase of software development that evaluates how efficiently an application performs under normal operating conditions. For the OptiCrop project, performance testing was conducted to verify that the application generates crop recommendations accurately, responds quickly to user requests, and provides a smooth user experience. The testing focused on validating the responsiveness of the web application, the reliability of the prediction model, and the overall stability of the system.

2. Objective

The objectives of performance testing are:

- To evaluate the response time of the application.
- To verify the accuracy and consistency of crop predictions.
- To ensure the application processes valid inputs without errors.
- To assess the stability of the web application during normal usage.
- To identify potential areas for performance improvement.

3. Testing Environment

The OptiCrop application was tested in the following development environment.

Component	Specification
Programming Language	Python
Web Framework	Flask
Machine Learning Library	Scikit-learn
Development Environment	Jupyter Notebook

Browser	Google Chrome / Microsoft Edge
Operating System	Windows / macOS

4. Test Scenarios

The following scenarios were considered during performance testing.

Test Scenario	Expected Result	Status
Application Launch	Application loads successfully	Passed
Home Page Loading	Home page loads without errors	Passed
Valid Input Submission	User inputs are accepted correctly	Passed
Crop Prediction	Recommended crop is generated successfully	Passed
Invalid Input Handling	Invalid or incomplete inputs are handled appropriately	Passed
Result Display	Predicted crop is displayed correctly	Passed

5. Performance Evaluation

The OptiCrop system demonstrated efficient performance during testing.

- The application successfully processed user requests without noticeable delays.
- Prediction results were generated within a few seconds for valid input values.
- Navigation between web pages was smooth and responsive.
- The machine learning model consistently produced crop recommendations based on the provided agricultural parameters.
- No critical runtime errors were observed during functional testing.

6. Model Performance

The trained Logistic Regression model achieved the following evaluation metrics:

Metric	Value
Accuracy	**97.27%**
Precision	**97.42%**
Recall	**97.27%**
F1-Score	**97.25%**
ROC-AUC	**0.9998**

These evaluation results demonstrate the effectiveness of the machine learning model in recommending suitable crops for the given agricultural conditions.

7. Observations

The following observations were made during testing:

- The web application responded efficiently to prediction requests.
- Input validation prevented incorrect data submission.
- The prediction model produced consistent results for valid inputs.
- The user interface remained responsive throughout testing.
- The overall system operated reliably under normal usage conditions.

8. Limitations

The current implementation has the following limitations:

- Performance evaluation was conducted using the available dataset and development environment.
- The application has not been tested under high concurrent user loads.
- Predictions depend on the quality and diversity of the training dataset.
- The current version focuses solely on crop recommendation.

9. Future Improvements

Future versions of the system can further improve performance by:

- Deploying the application on cloud infrastructure.
- Optimizing prediction latency for larger datasets.
- Supporting concurrent users through scalable deployment.
- Implementing caching mechanisms for repeated requests.
- Integrating real-time weather and soil data services.

10. Conclusion

Performance testing confirmed that the OptiCrop system operates efficiently and reliably under normal usage conditions. The Flask web application successfully integrates with the trained machine learning model to generate accurate crop recommendations with minimal response time. The strong evaluation metrics and stable application performance demonstrate that the system is suitable for assisting users in making informed agricultural decisions while providing a solid foundation for future enhancements.